UCH402: HEAT TRANSFER

L	T	P	Cr
3	1	2	4.5

Course Objective:
To understand the fundamentals of heat transfer mechanisms in fluids and solids and their applications in various heat transfer equipment in process industries.

Heat Transfer: Introduction, Applications, Relation between heat transfer and thermodynamics, Transport properties, Heat transfer coefficients.

Conduction: Fourier's law, Thermal conductivity, Heat conduction equations: Rectangular, cylindrical and spherical coordinates, Composite wall structure, Insulation and its optimumthickness, Extended surfaces, Unsteady state conduction.

Convection: Newton's law of cooling, Boundary layer theory, Heat transfer in laminar and turbulent flows inside tubes, Colburn analogy, Heat transfer by external flows across: Cylinders, tube bank and spheres, Natural convection, Convection with phase change: Boiling and condensation.

Radiation: Basic equations, Emissivity, Absorption, Black and gray body, Thermal radiation between two surfaces.

Heat Exchangers: Classification of heat exchangers, LMTD and□ -NTU methods, Heat exchangers: Double pipe, shell and tube, air-cooled, plate type, Fouling, Concept of pinch technology and its application.

Evaporators: Classification, Single and multiple effect evaporators, Enthalpy balance, Performance of evaporators: Capacity and economy, Methods of feeding.

Laboratory Work:
Thermal conductivity of a metal rod, Thermal conductivity of insulating power, Emissivity measurement,Parallel flow/counter flow heat exchanger, Heat transfer through composite wall, Drop wise & film wisecondensation, Heat transfer through a pin-fin, Heat transfer in natural convection, Heat transfer in forcedconvection, Critical heat flux, Stefen-Boltzman's law of radiation, Heat flow through lagged pipe, Shell and tube heat exchanger.

Course Learning Outcomes (CLO)
Upon completion of the course, students will be able to:
1. solve conduction, convection and radiation problems
2. design and analyse the performance of heat exchangers
3. analyse the steam economy and performance of evaporators

***Text Books*:**
1. *McCabe, W.L., Smith J.C., and Harriott, P., Unit Operations of Chemical Engineering, McGraw-Hill (2005).*
2. *Holman, J.P., Heat Transfer, Tata McGraw-Hill Education (2008).*

3. *Sinnott Ray and Towler Gavin, Coulson and Richardson's Chemical Engineering series Chemical Engineering Design Volume 6, 5th edition (2013).*

***Reference Books*:**

1. *Kern, D.Q., Process Heat Transfer, Tata McGraw-Hill (2008).*
2. *Frank, P.I. and David, P.D., Fundamentals of Heat and Mass Transfer, John Wiley & Sons (2007).*
3. *Cengel, Y.A., Heat and Mass Transfer, Tata McGraw-Hill Publishing Company Limited (2007).*
4. *Alan, S.F., Leonard, A.W. and Curtis, W.C., Principles of Unit Operations, Wiley India (P) Ltd., (2008).*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	45
3	Sessional (may include Quizzes/Assignments/Tutorials/Lab evaluations)	30